Binary addition

1) Half adder (Performs addition of two bits)

x	y	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

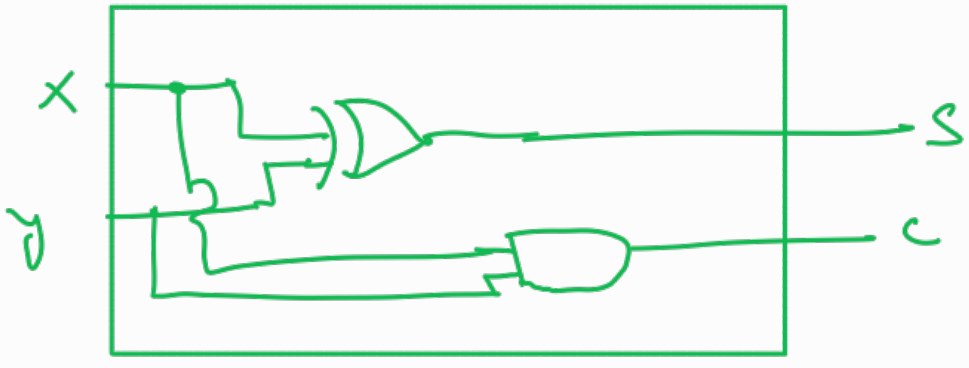
Sop eq

T·T (output value → 1)

$S = \bar{x} \cdot y + x \cdot \bar{y}$

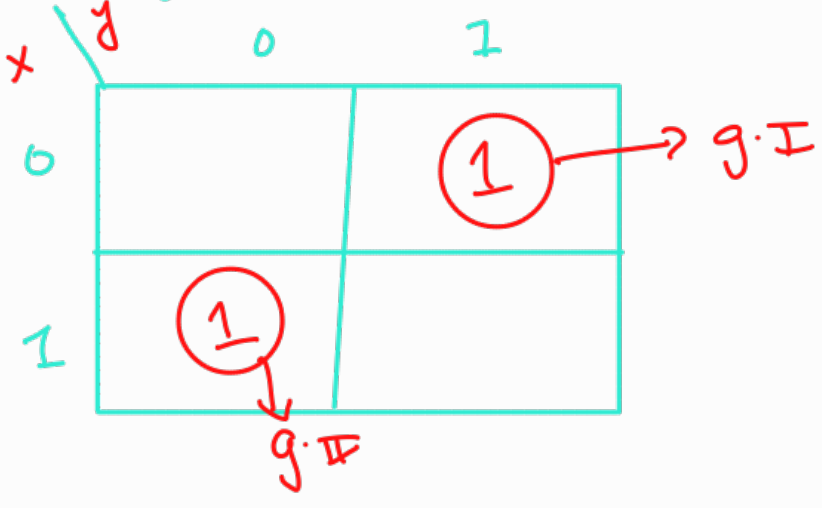
$= x \oplus y$

$C = xy$



2^2 = 4 cells

Deriving equations from k-map.



g·I

x=0 y=1

$\bar{x}y$

g·II

x=1 y=0

$x\bar{y}$

$S = \bar{x}y + x\bar{y}$

Full Adder:: Adds three input bits only.

x	y	z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sop eq for Sum

$$S = \bar{x}\cdot\bar{y}\cdot z + \bar{x}\,y\cdot\bar{z} + x\bar{y}\bar{z} + xyz$$

$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

$$C = \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$$

$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

$$S = x\bar{y}z + x\bar{y}\bar{z} + \bar{x}y\bar{z} + xyz$$

$$S = \bar{y}(\bar{x}z + x\bar{z}) + y(\bar{x}\bar{z} + xz)$$

$$S = \bar{y}(\bar{x}z + x\bar{z}) + y$$

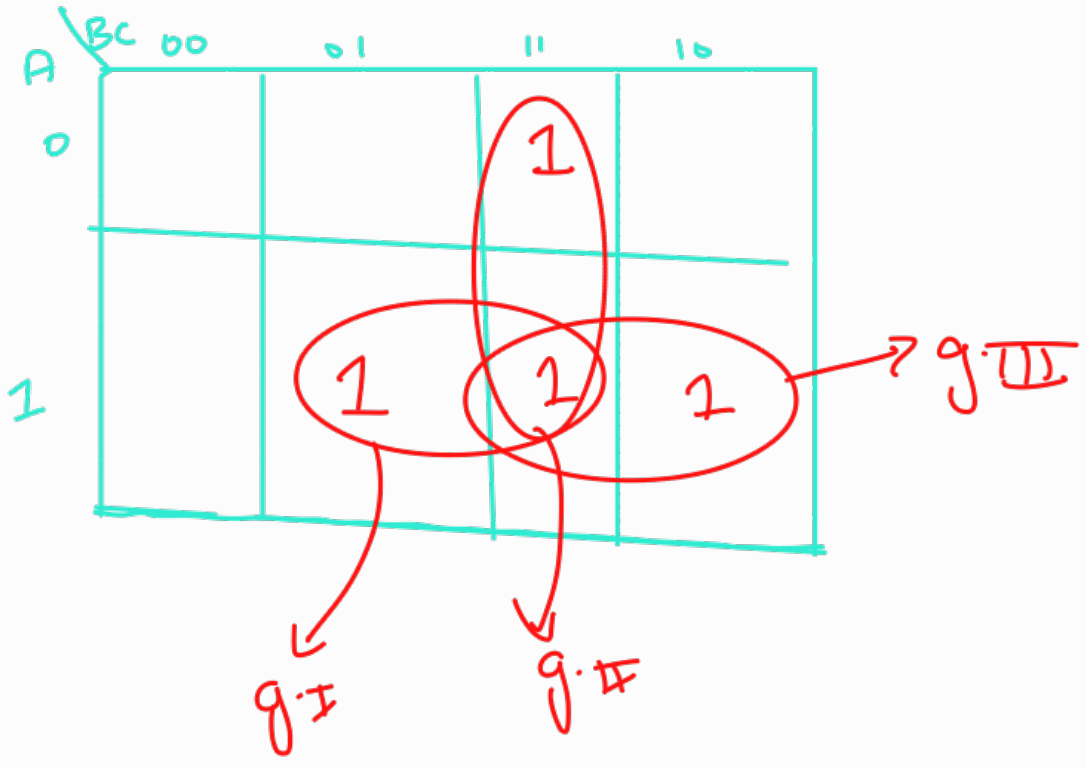
XOR eq

$$A \oplus B = (A\cdot\bar{B}) + (\bar{A}\cdot B)$$

$$\boxed{S = \bar{y}(x \oplus z) + y}$$

k-map will be of $2^3 = 8$ cells.

Using k-maps to find carry equation.



g.I x = 1 y z

y	z
0	1
1	1

(x)

x.z

g.II

x = 0 then x = 1 |Discord

y z

1 1

→ y.z

g.III

x = 1

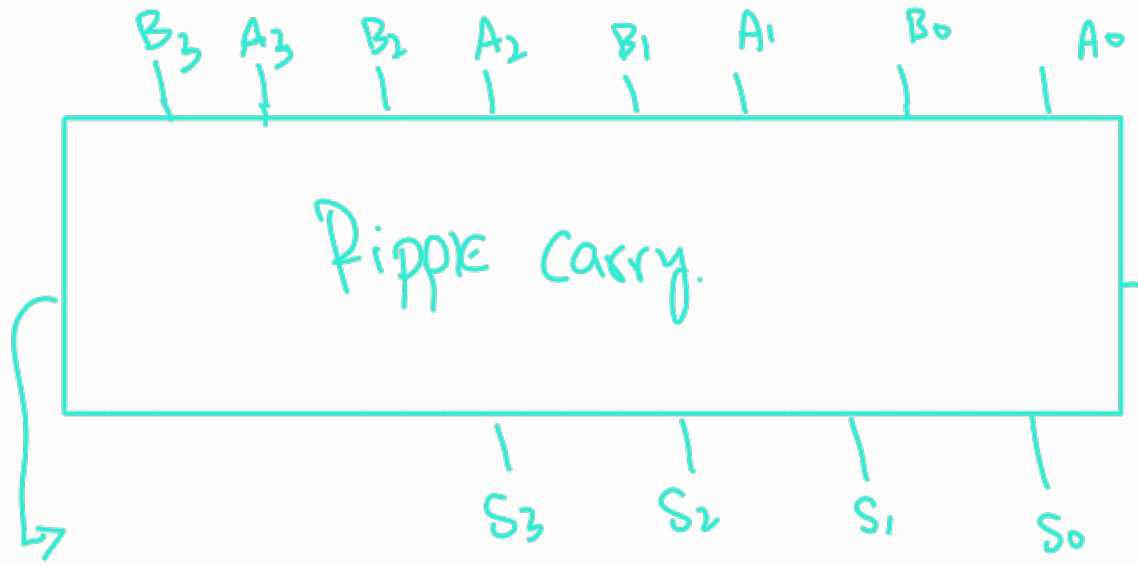
y	z
1	1
1	0

(x)

x.y

C = xy + yz + xz

4bit ripple carry adder.



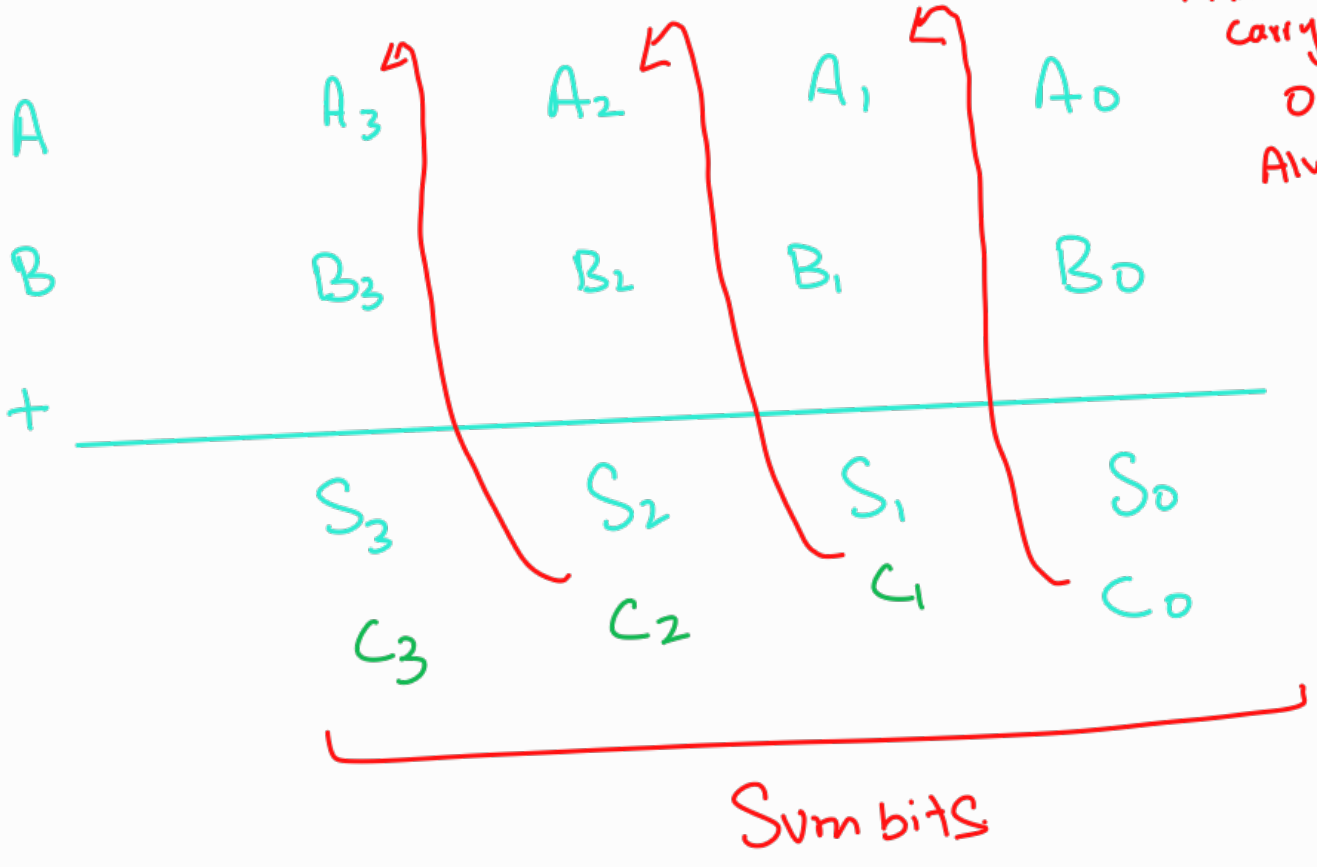
Two numbers A and B.

Both have 4 Bits

4 bits + 4bits no = 4bits Sum

Carry generate Possibility

Intial carry is 0 Always



BCD to excess 3 convertor

A B C D	w	x	y	z
0 0 0 0	0	0	1	1
0 0 0 1	0	1	0	0
0 0 1 0	0	1	0	1
0 0 1 1	0	1	1	0
0 1 0 0	0	1	1	1
0 1 0 1	1	0	0	0
0 1 1 0	1	0	0	1
0 1 1 1	1	0	1	0
1 0 0 0	1	0	1	1
1 0 0 1	1	1	0	0

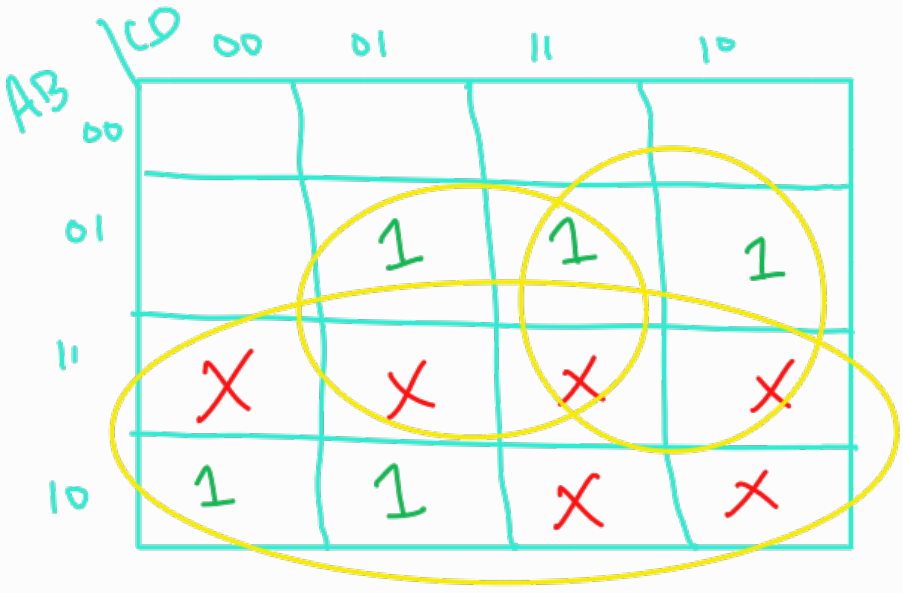
XS3 is given when we add 3 in BCD code

Making equation using k-map.

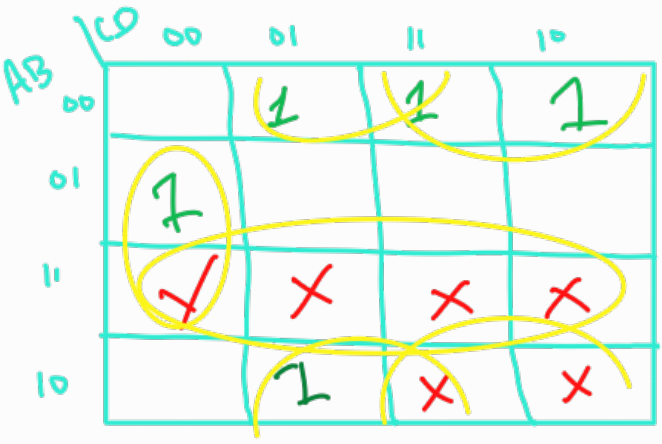
We will have to make 4 Seperale k-maps for each w x y z each having

16 cells as no. of input is 4. $2^4 = 16$ cells.

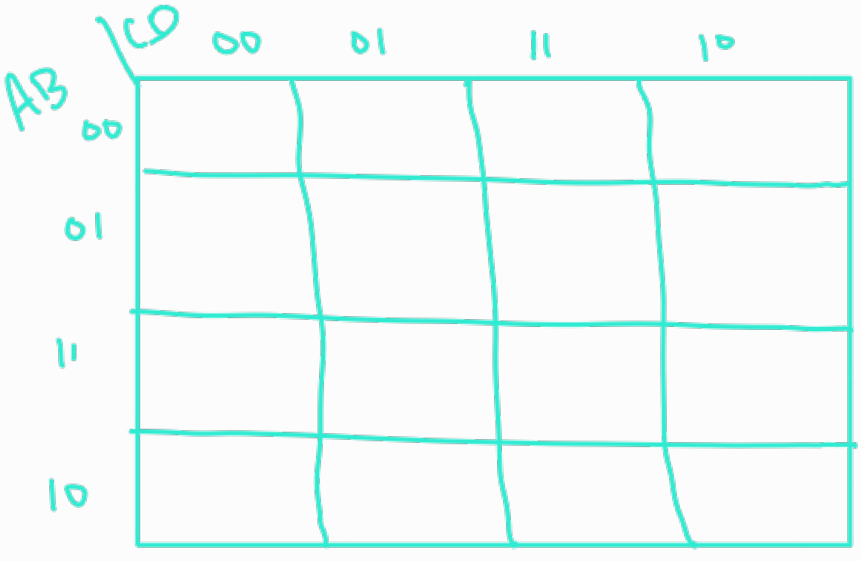
W


CD
00
01
11
10
AB
00
01
11
10
1
1
1
X
X
X
X
1
1
X
X

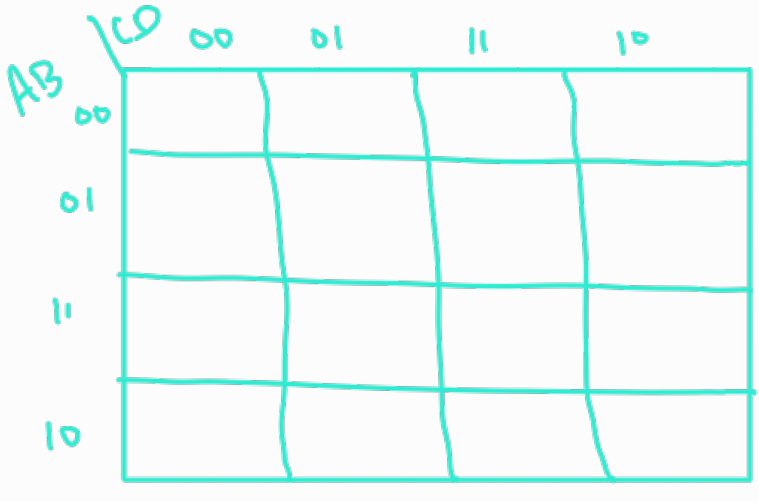
X


CD
00
01
11
10
AB
00
01
11
10
1
1
1
1
1
X
X
X
1
X
X

Y


CD
00
01
11
10
AB
00
01
11
10

Z


CD
00
01
11
10
AB
00
01
11
10

half Subtractor → meaning we have 2 inputs and two outputs

x	y	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

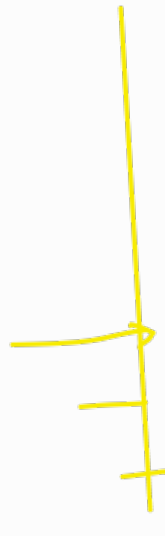
10
- 1
1

equations $D = \bar{x}y + x\bar{y}$ $B = \bar{x}y$

$D = X \oplus y$

full subtractor : we need 3 bits (Inputs)

X	y	z	D	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



Making output equations:-

$$D = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

$$B = \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$$

Full 4 bit adder example:-

A 1 1 0 1

B 1 0 1 1

$A_3\ A_2\ A_1\ A_0$

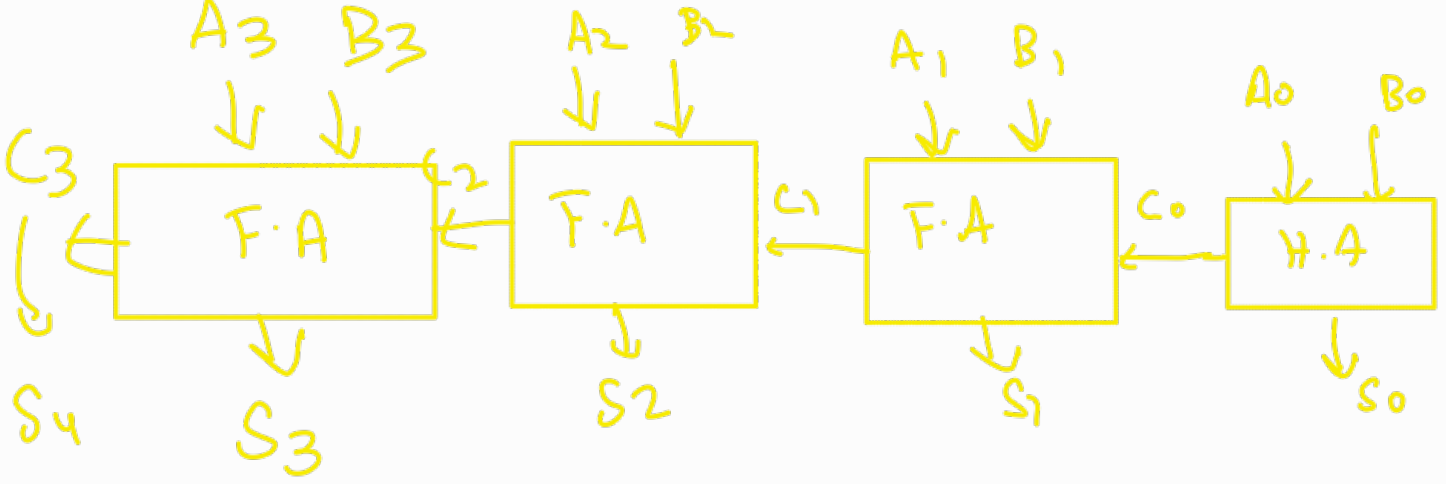
A → 1 1 0 1

B → 1 0 1 1

$B_3\ B_2\ B_1\ B_0$

$S_3\ S_2\ S_1\ S_0$

first we are adding $A_0 + B_0$ here using half adder.



Adder | Subtractor in one circuit.

XOR Operation

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

$$x \oplus 1 = \overline{x}$$ → XOR operation



(Implementation of 1's complement)